

Modeling Selection in Active Cross-situational Word Learning

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Abstract

Word learning is a complex process, influenced not only by the words and referents in the language environment, but also by what learners selectively attend to, which in turn is likely determined by their prior knowledge.

Keywords: active learning; cross-situational word learning; self-directed learning; uncertainty reduction

Introduction

Early word learning is often thought of as largely determined by the learners' linguistic environment, where they can only learn what is present in the input. Increasingly, research is finding that learners—even at a young age—are involved in selecting and shaping their input, and that such self-directed, active learning can accelerate acquisition. Understanding the strategies that learners of all ages use to shape their experience will help us build and improve theories of language acquisition. It can also help us understand what cognitive mechanisms and strategies learners are able to deploy, and consequently enable us to design curricula that better match their expanding capabilities. For example, although the classic conceptual @hunter1988 model posits that infants have preferences for both novel and familiar stimuli, and novelty preferences have been implicated in children's word learning at some ages [e.g. 22-month-olds: @mather2012; @horst2011], a recent study found that infants do not show a bias to attend to uncertain/novel stimuli (find citation on langcog slack..).

Although numerous computational models of word learning have been proposed, the experimental paradigms they are tested on tend to be passive cross-situational word learning studies, in which participants are exposed to a series of ambiguous training trials, each of which presents multiple word-referent pairs, but with no indication of which word refers to each referent. While model comparisons based on these passive learning paradigms are suggestive (although a full overview is not possible here), applying these models to help understand active learning paradigms – where learners select their input at some stage (e.g., requesting a label for a particular referent) can in theory be more diagnostic of the learning mechanisms at play, and may additionally help explain individual differences in learning. And yet, few active word learning experiments have been conducted, and fewer still have attempted to understand active learning with computational models [@Kachergis2013].

The goal of our current study is to test how well previously proposed models of cross-situational word learning can explain the observed results of active learning experiments from @Zettersten2021, which found that adults – but not children – were likely to select referents with ambiguous labels. Aside from assessing the ability of the models to account for the overall data, it is of particular interest whether particular model parameters (e.g., learning rate, memory decay rate, and novelty vs. familiarity bias) are associated with individuals' 1) learning performance and 2) deployed active learning strategy. We will also investigate age-related changes in model parameters.

Past work has shown that both children and adults are active word learners, systematically seeking new information word meanings [@hembacher_childrens_2020; @janakieski_gaining_2022; @Zettersten2021; other refs], with important implications for understanding how we are able to quickly build rich vocabularies [@Hidaka2017]. Yet what types of active learners are children about word meanings, exactly? While work has begun to characterize a variety of strategies that learners pursue when encountering novel words [@Kachergis2013; @zettersten_active_2023], we lack a formal understanding of the specific sampling strategies that word learners use [@Settles2012; though see @gelderloos_active_2020; @Keijser2019]. Recent work has documented broad changes in the sampling strategies that children and adolescents deploy across development across several cognitive domains, e.g. causal learning tasks [@nussenbaum_causal_2019] and foraging tasks [@nussenbaum_novelty_2023; @meder_directed_2021]. In the current work, we therefore explore what sampling decision rules best explain both children's and adults' information-seeking behavior when learning novel word-referent associations.

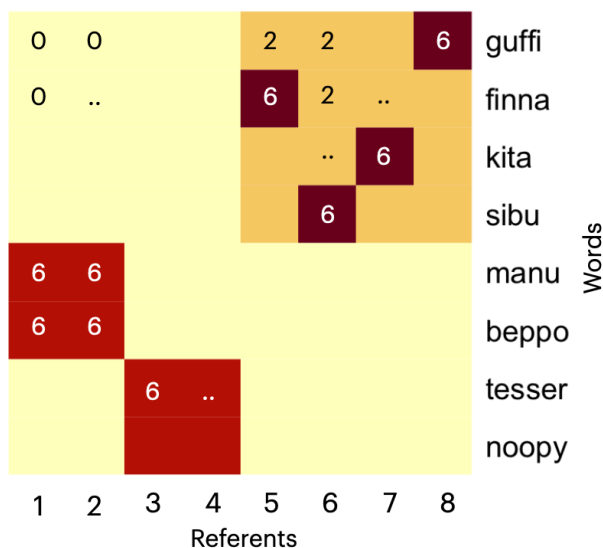


Figure 1: Word-referent co-occurrence matrix for Study (Experiment 1, Zettersten and Saffran, 2020).

GK: we can also test this based on best-fitting model parameters: “Intriguingly, we found that participants’ sampling behavior was correlated with their test accuracy: participants who chose more ambiguous items during the Sampling Phase also more accurately identified object-label associations. However, the directionality of this correlation is unclear. One possibility is that learners were more accurate on the test because they sampled more ambiguous items. Another possibility is that success fully learning novel words during training makes it more likely that learners will subsequently focus on ambiguous items during the Sampling Phase. In the supplementary in-lab replication experiment (see S1 for further details), we found preliminary evidence supporting the latter explanation— as participants become more successful at learning the novel label-object associations, they also become more likely to seek out information about ambiguous items.”

kids weren’t more likely to sample the ambiguous referents: either they are hypothesis testers (and saw no disconfirming evidence), or maybe they have less of a focus on uncertainty (and perhaps slower learning rate..and how does memory influence sampling choice? recency bias if worse memory?)

Method

Data

All data were taken from @Zettersten2021. Data for our Study 1 were pooled from Experiment 1 (31 adults), Experiment 2 (40 children, 4-9 years of age), and adults in the Ambiguous Condition of Experiment S1 (58 adults), which had the same design as Experiments 1 and 2, except for an additional test before the selection phase.

The training phase of Study 1 contained 24 FixMe-second training trials, each comprised of 2 referents and 2 words. The particular words and referents involved in each pairing were randomized per participant, but the same training trial sequence—and thus co-occurrence statistics—were experienced by all participants. The accumulated word-referent co-occurrences are shown in Figure 1.

Data for Study 2 come from Experiment 3, comprising 58 children, 3 to 8 years of age.

Selection trials for kids differ slightly from adults in that there was no random label-referent pair given: thus, the evidence children received after selection was unambiguous, while adults received a training trial with a randomly-selected label and its corresponding referent.

Models

We focused on analyzing data from the familiarity- and uncertainty-biased associative model [Kachergis2012, @Kachergis2017], and variations of that model, described below. We chose not to test any hypothesis-testing (HT) model as they are not consistent with the adult data in @Zettersten2021: for ambiguous items in the training phase, there was never any disconfirming evidence, so a hypothesis-tester’s initial guess (i.e. hypothesis) for each label would be confirmed on subsequent appearances, and there would be no reason for (or basis for) preferentially selecting ambiguous referents during the sampling phase, which is what adults were observed to do. While it is possible that children – who were at chance for selecting ambiguous vs. unambiguous referents – are employing a hypothesis-testing strategy, some of the selection strategies we want to evaluate across development are only possible to use if a learner is simultaneously entertaining multiple hypotheses for a given label – equivalent to associations rather than the mutually-exclusive hypotheses favored in HT accounts [e.g. @woodard2016].

Familiarity- and Uncertainty-biased Associative Model (FUAM) The FUAM assumes that learners accumulate associations between all presented words and referents on each trial, but that their attention is biased more towards particular word-referent associations on the basis of prior experience. In particular, more attention (i.e. associative weight) is directed to associations between words and referents that have appeared together on earlier trials (i.e., a familiarity bias). In addition, more attention is directed to stimuli that have higher uncertainty (i.e. entropy, H) in their associations, whether because they are novel (and thus have no pre-existing associations) or because they have diffuse associations with several different stimuli. The model assumes learners knowledge grows trial-by-trial, and that a fixed amount of associative weight (learning rate χ) is distributed among the presented word-referent associations on a given trial in proportion to the prior familiarity of the association and the uncertainty of the involved word and referent. The relative strength of the familiarity and uncertainty bias is governed by a parameter λ :

for values of $\lambda > 1$ the model focuses more attention on uncertain stimuli, while for values of $\lambda < 1$ the model focuses more attention on familiar word-referent associations. Associations are also assumed to decay from trial to trial, at a rate governed by memory fidelity parameter α . The update rule for adjusting the association $M_{w,o}$ between word w and referent o on a given trial is

$$M_{w,o} = \alpha M_{w,o} + \frac{\chi \cdot e^{\lambda \cdot (H(w) + H(o))} \cdot M_{w,o}}{\sum_{w \in W} \sum_{o \in O} e^{\lambda \cdot (H(w) + H(o))} \cdot M_{w,o}} \quad (1)$$

When tested on word w , the model assumes learners choose referent o from the offered alternatives in proportion to their associative strength $M_{w,o}$ (i.e., Luce choice).

The other models we tested are variations on this model.

Familiarity-biased Associative Model (FAM) The familiarity-biased associative model (FAM) implemented just the familiarity bias of the FUAM, without the uncertainty bias term, and thus only implements a bias for already-strong associations, using learning rate (χ) and decay α parameters.

Uncertainty-biased Associative Model (UAM) The uncertainty-biased associative model (UAM) implemented only the uncertainty bias of the FUAM, without the familiarity bias term, and uses the same three parameters as the FUAM.

Novelty-biased Associative Model (NAM) The novelty-biased associative model (NAM), like the UAM, has no familiarity bias, but instead of the entropy terms ($H(w)$ and $H(o)$) uses novelty (e.g., $1/(\text{frequency}(w) + 1)$).

Fitting Procedure

We first fitted each model to each participant’s data individually, finding parameter values per participant to maximize the log-likelihood of the model’s choices with respect to theirs. We fitted each model to just the passive training trials, as well as to the passive training trials and the participant’s selection trials, to confirm that including the selection trials yielded a better fit. We will report each model’s average fit, and select the best-fitting model overall for further analysis.

Modeling Sampling Choices

Next, we used the best-fit models of participants’ learning to investigate their selection strategies in the sampling phase of the experiment. We considered three sampling decision rules commonly attested in the information-seeking literature to explain active choices: maximizing entropy reduction (information gain), maximizing the probability associated with labels (probability gain), and margin sampling.

Maximizing entropy reduction. In this decision rule, the model’s goal is to maximally reduce the entropy of the object-label probability distribution across objects [Settles2012; Nelson2005]. Sampling choices are modeled based on comparing the relative entropy reduction provided

by making a given sampling selection of a referent s_o (across the labels w):

$$\max_{s_o, w} (H(w) - H(w|s_o)) \quad (2)$$

Modeling selections were made by determining, on any given trial for a participant, which object maximally reduced entropy (averaging across all possible random distractors that could be chosen in the case of the adult data).

Margin sampling. An alternative strategy documented in describing active sampling strategies is focusing on reducing uncertainty for a more constrained set of items [markant_self-directed_2016]. This strategy can be formalized as *margin sampling*, in which the model focuses only on the boundary between two alternative options. Margin sampling predicts that learners will focus on instances where there is highest ambiguity between the two most likely labels for a given referent. Formally, if the distribution of labels ($\{l_1, l_2, \dots, l_n\}$) for a given referent o are ordered from most (p_1) to least (p_n) likely, $\{p_1, p_2, \dots, p_n\}$, the margin between labels is given by

$$LM(x) = 1 - (p_1 - p_2) \quad (3)$$

In other words, the model selects the object that has the greatest ambiguity or competition between its two strongest associates.

Maximizing probability gain. In this decision rule, the model’s goal is to maximally increase the probability associated between a random label and its object [nelson_experience_2010]. Selection choices are modeled based on determining the increase in (correct) association strength provided by a each potential object selection. The final object was selected that led to the largest increase in association strength (averaging across all possible distractors that could be chosen in the case of the adult data).

$$\max_{s_o, w} (M_{w,s_o}) - \max_{w, o} (M_{w,o}) \quad (4)$$

Alternative strategies. We also explored several additional strategies. First, we also considered a *maximum certainty* (or confirmatory) strategy. Learners may gravitate towards confirming choices they already have high certainty for [Watson1960; Zettersten_active_2023]. Second, we considered a *novelty-based* strategy in which the model based its choices on the relative novelty (or familiarity) of the choice options. Third, we consider a random selection strategy as a baseline comparison.

Results

Model	N best fit	strength	uncertainty	novelty	FUAM
FUAM	67	1.66	1.66	1.65	1.42
strength	24	2.00	2.09	2.00	2.00
novelty	3	3.22	3.42	3.01	3.19
uncertainty	96	3.56	2.57	3.49	3.37

For all four models, including the selection trials yielded a better fit to the data, showing that it is not unreasonable to apply these models to the selection trials, and supporting the idea that the model’s knowledge state and parameters may bear some relation to participants’ selection strategies. For model comparison, we calculated BIC (Bayesian Information Criterion), defined as $BIC = \ln(N)k - 2\ln(L)$, where L is the maximized likelihood, k is the number of parameters, and N is the number of data points (here, $N = 1712$ test trials). The model with the lowest BIC is selected, as it balances model complexity and fit to the data. The strength-biased model yielded $BIC = 1036.17$, the novelty-biased model scored $BIC = 1026.68$, and the familiarity- and uncertainty-biased model (FUAM) had $BIC = 975.22$. The uncertainty-biased model (UAM) yielded the best fit, with $BIC = 859.09$. Recognizing that different participants may be operating under different learning regimes, we opted to investigate how many participants were best fitted by each model. The uncertainty-biased model best matched behavior of 96 of the 190 participants (51% of the sample), and the average fit (negative log-likelihood; see Table 1) of the model to its best-fitted participants was much better (i.e. lower) than any other model (2.57 vs. 3.37 by the FUAM model). The FUAM achieved the best fit for 67 participants, with the best average fit to that group seen in Table 1 (1.42). The FUAM also matched the best fit of the strength-biased model to 3 decimal places (see Table 1) for the 24 participants that were best fitted by that model, making for a total of 91 participants (48% of the sample) that are best or nearly equally well accounted for by the FUAM model (the uncertainty model did not fit as well to the strength participants). The novelty-biased model best fitted only 3 participants, although it’s worth noting that its average fit to the strength participants was nearly the same as the FUAM and strength models. We will focus our remaining analyses on the model parameters for the two best-fitting models, the familiarity- and uncertainty-biased associative model (FUAM) and its cousin the uncertainty-biased associative model (UAM).

Model parameters

The FUAM and UAM models achieved similar fit ($|NLL_{FUAM} - NLL_{UAM}| \leq 0.1$) for 98 participants. The UAM achieved better fit than the FUAM ($NLL_{FUAM} - NLL_{UAM} > 0.1$) for 56 participants, and the FUAM achieved better fit than the UAM for the remaining 36 participants. We will call these participants the *Either*, *UAM*, and *FUAM* groups, respectively.

Group	N	chi	lambda	alpha	Mean accuracy
Either	98	0.32	5.34	0.77	0.59
FUAM	36	0.41	4.97	0.77	0.65
UAM	56	0.26	8.76	0.69	0.57

Table 2 shows the average parameter values for each group of participants, and the mean accuracy and number of participants per group. None of these values are strikingly different

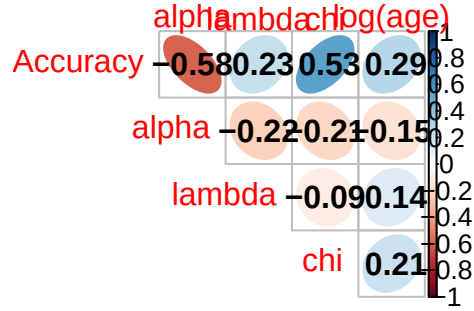


Figure 2: Correlation of best-fitting model parameters

than the overall average parameter values (mean $\chi = 0.32$, mean $\lambda = 6.28$, mean $\alpha = 0.75$), though the UAM group shows lower learning rate ($\chi = 0.26$), lower memory fidelity ($\alpha = 0.69$), and a higher focus on uncertainty ($\lambda = 8.76$). The FUAM group has a somewhat higher learning rate ($\chi = 0.41$) and the greatest overall accuracy ($M_{acc} = 0.65$).

Figure 1 shows the correlations between participants’ accuracy, best-fitting model parameters, and log(age). All correlations are significant (e.g., λ vs log(age) $r = 0.14$, $t(187) = 1.99$, $p < .05$) except λ vs. χ . Most notable is that accuracy is strongly related to higher learning rate χ ($r = 0.53$) and lower memory fidelity ($\alpha = 0.58$; better learning is achieved by forgetting early, ambiguous evidence), and only weakly related to λ ($r = 0.23$) and age ($r = 0.29$). A targeted analysis found that children (age < 18) are more likely than adults to be familiarity-focused ($\lambda < 1$) rather than uncertainty-focused ($\chi^2 = 7.41$): of 22% of adults had $\lambda < 1$, while 41% of children did.

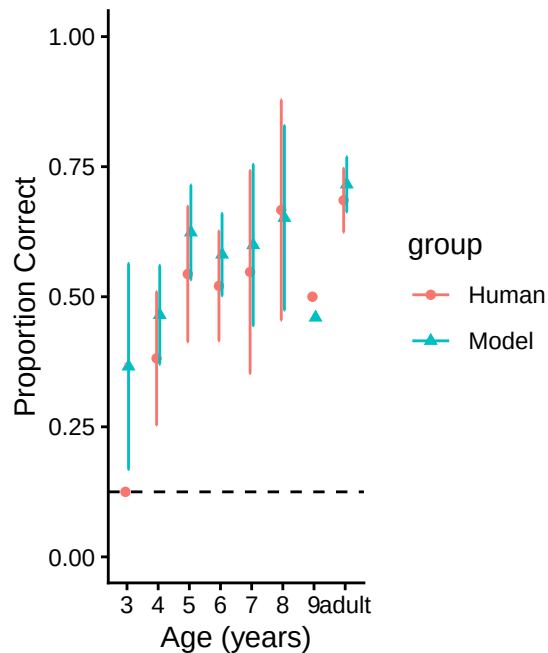


Figure 3: Model vs. human performance by age.

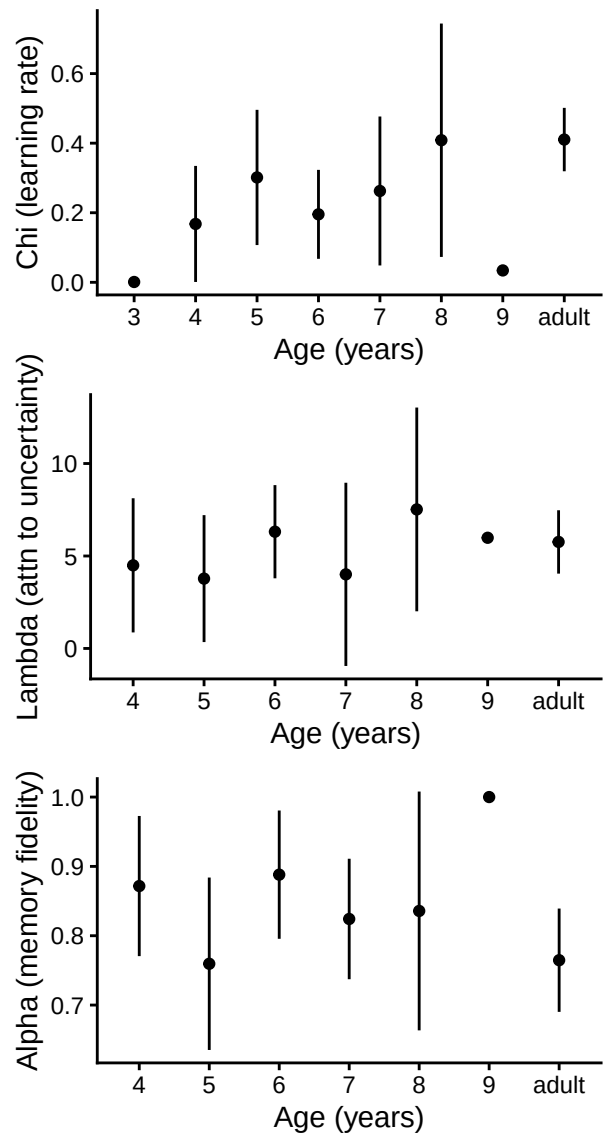


Figure 4: One column image.

Selection Strategies

The most likely selection strategy for each participant was calculated. Confirmatory was the most common strategy (42% of participants), followed by entropy (29%), margin sampling (18%), random (8%), and novelty (3%). Adults were more likely to use entropy maximization (39% vs. 20% of children), whereas children were more likely to use a confirmatory strategy (62% vs. 22% of adults). Margin sampling was used at a similar rate by children (14%) and adults (22%), and a random selection was most likely for 13% of adults, but only 3% of children. Novelty was the most likely strategy for 5% of adults, and no children.

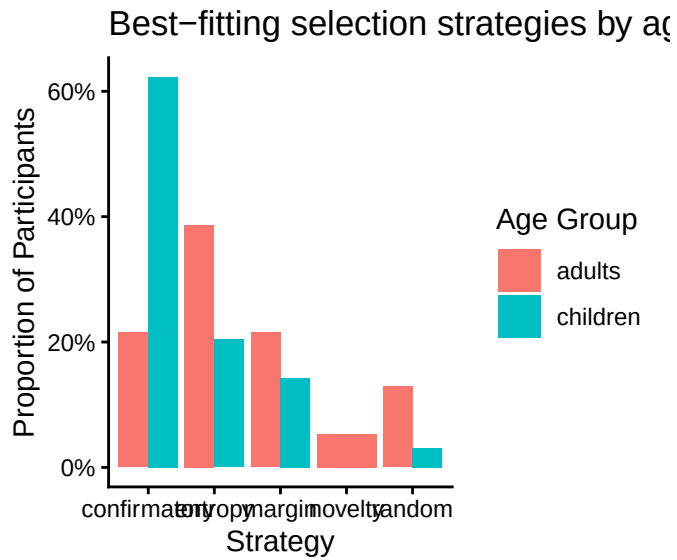


Figure 5: Best-fitting selection strategies by age group.

strategy	n	proportion
confirmatory	81	0.42
entropy	56	0.29
margin	34	0.18
random	15	0.08
novelty	5	0.03

Discussion

Acknowledgements

Place acknowledgments (including funding information) in a section at the end of the paper.

References